

Enhancing Ad Effectiveness: A Randomized Experiment on Specific vs. General Fundraising Appeals

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2025-04-25

Abstract

We conducted a clustered, between-subjects field experiment in partnership with the South Yuba River Citizens League to evaluate whether quantifiable messaging increases donor engagement in digital fundraising ads. Sixty ZIP codes—stratified by past donation levels and balanced on median income and river proximity—were randomly assigned to receive either a treatment ad with specific quantifiable outcomes or a control ad conveying only general support. Ads were served via Meta Ads on Facebook and Instagram over one week, and click-through rate (CTR) was recorded as the primary outcome. During delivery, we observed systematic covariate imbalance across age, device, and placement categories driven by Meta’s targeting algorithm despite clustering on ZIP code. We attempted to adjust for these imbalances by including available covariates (age, gender, device, placement) in our regression models. Regression results revealed no statistically significant difference in CTR between treatment and control. We discuss how platform-induced imbalances and data constraints inform future designs.

1 Introduction: Importance and Context

Strategic allocation of funds is essential for maximizing impact while ensuring financial stability. This is especially true for nonprofit organizations, where directing money effectively demonstrates accountability to donors and builds trust that their contributions are being used for causes they care about (Hattaway Communications, 2023). Given limited resources, nonprofits must prioritize spending to make the most of every dollar.

Research supports the importance of transparency in charitable giving. A study titled “The Donor Is in the Details” by Cryder et al. found that providing tangible, specific information about how donations are used significantly increases generosity. Participants who received detailed information about a charity donated 80–85% more than those who received only general information.

Our research builds on these concepts to test the effectiveness of messaging strategies on donor engagement. Specifically, our experiment tests the question:

Does providing potential donors with specific, quantifiable outcomes of an organization’s initiatives increase donor engagement compared to providing only general information about the organization’s mission?

As our research centers upon the effectiveness of messaging strategies in donation requests and advertisements, we partnered with a nonprofit organization to execute our experiment. Our partner organization, the South Yuba River Citizens League (SYRCL), is a nonprofit organization in Northern California working on efforts to protect and restore the waters, forests, and communities of the Yuba River watershed.

2 Experiment Design

We designed a geo-based digital ad experiment using geofencing to measure the causal impact of quantifiable organizational outcomes in donation advertisements on donation behavior. Our experiment had one treatment advertisement and one control advertisement, and the only difference between the two advertisements was the messaging. The treatment advertisement’s messaging included quantifiable organizational outcomes, while the control group’s messaging only included general information about the organization’s mission. We used Meta Ads to distribute our advertisements, and employed geofencing to eliminate algorithmic biases from the platform. We distributed the ads over a period of one week for both treatment and control groups. Our experimental design is a between-subject design, where each participant either saw an ad with quantitative data on the outcomes of a donation (treatment) or only general information (control).

To measure our outcome variable donation behavior, we tracked clickthrough rates (CTR) on each ad. Using the Meta Ads platform, we also collected additional covariates including gender, age, device type, and placement.

2.1 Comparison of Potential Outcomes

The potential outcomes of this experiment could be one of three things: 1. The specified, quantifiable advertisement motivates individuals to have more propensity to click on the ad and give to the charity 2. The specific messaging has no effect on the way an individual gives to the charity 3. The more general ad leads people to have a higher tendency to donate to the charity. Considering the prior research and experiments, the expectation of the potential outcome is to see a statistically significant increase in CTR when the quantifiable and specific messaging is delivered to the Meta users.

2.2 Randomization Process

To assign ZIP codes to treatment and control groups, we first identified the 187 ZIP codes that had donated to SYRCL in the past two years and selected the top 60 by total donation amount. These 60 ZIP codes were then stratified into three blocks—high, medium, and low donors—based on their prior contribution levels. Within each block, we performed a covariate balance check on two key variables: ZIP code-level average income (from California state data) and proximity to the Yuba River (calculated using river geocoordinates). Finally, ZIP codes in each block were randomly allocated to treatment or control, ensuring balanced comparison groups.

2.3 Treatment

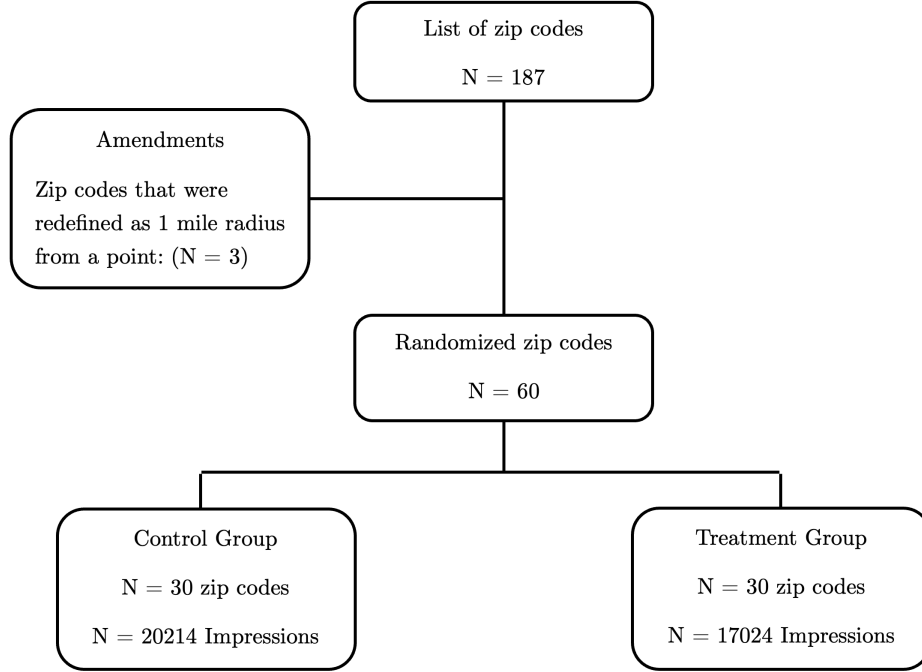
To isolate the effect of quantifiable messaging on donor behavior, we developed two ad variants in collaboration with the South Yuba River Citizens League (SYRCL) that shared identical imagery, layout, visual branding, platform placement, and call-to-action (“Give today”), differing only in textual framing. The treatment ad included the explicit impact statement: *“Your \$50 donation lets us teach one additional student about rivers, streams, and wildlife in their community.”* The control ad read: *“Help support our efforts to teach more students about rivers, streams, and wildlife,”* omitting any numerical outcome.

This design is grounded in evidence that donors respond more strongly when the tangible results of their gift are quantified (Cryder et al., 2013). By holding all other ad features constant, we ensure that any difference in click-through rate can be causally attributed to the specificity of the messaging alone.

Ads were served via Meta Ads on Facebook and Instagram, with randomization at the ZIP code level so that every resident within a given ZIP code saw the same ad version. This was also so that the same Meta targeting algorithm would be applied to all units. We classify this experiment as a between-subjects design.

2.4 CONSORT

Below is the CONSORT diagram of this experiment.



2.5 Power Calculation

To determine the appropriate sample size for our experiment, we conducted a power analysis that accounts for the clustered design of our study—where treatment is assigned at the ZIP code level. This analysis estimates the number of ZIP codes required to detect a statistically significant difference in donation behavior between our treatment and control groups.

Motivation and Theoretical Basis

Our treatment design builds upon the experimental findings of Cryder et al. (2013), which demonstrated that providing detailed, outcome-specific information can increase charitable giving by over 80

While this result is promising, our field-based study differs in important ways: (1) participants will not be aware that they are part of an experiment, (2) there is no participation incentive, and (3) donations will be self-initiated based on passive exposure to Meta Ads. For these reasons, we included more conservative treatment effects in our simulations.

Simulation Design

We simulated a hypothetical population across 200 ZIP codes, stratified into high-, medium-, and low-income areas. Each ZIP code contained 500 individuals, with income distributions and baseline donation behavior modeled using log-normal distributions that reflect empirically plausible variance:

- High-income ZIPs: Mean \$50 donation, SD \$200
- Mid-income ZIPs: Mean \$30 donation, SD \$120
- Low-income ZIPs: Mean \$10 donation, SD \$40

Treatment was applied at the ZIP code level. For each simulation, we:

1. Sampled a fixed number of ZIP codes (12 to 120, in increments of 6)
2. Randomly assigned ZIPs to treatment or control within each income stratum

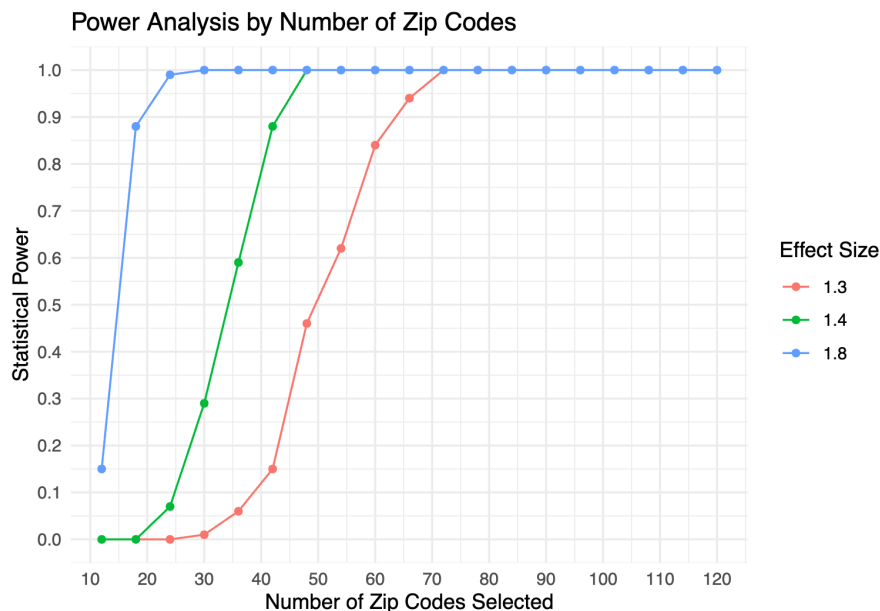
3. Simulated a treatment effect by scaling donation outcomes in the treatment group by a multiplicative factor (1.3, 1.4, or 1.8)
4. Estimated the treatment effect using OLS with cluster-robust standard errors at the ZIP level
5. Recorded the proportion of simulations (out of 100) in which the treatment coefficient was statistically significant at $p < 0.05$

Results and Interpretation

The simulation results indicate the number of ZIP codes required to achieve 80% power under varying effect sizes:

- **Effect Size 1.8 (80% increase):** ~20 ZIP codes
- **Effect Size 1.4 (40% increase):** ~40 ZIP codes
- **Effect Size 1.3 (30% increase):** ~60 ZIP codes

Based on these findings, we selected a sample size of 60 ZIP codes—30 assigned to treatment and 30 to control. This ensures we are powered to detect moderate effect sizes of around 30% or higher, which are grounded in theory but adjusted for real-world noise.



3 Analysis

3.1 Covariate Balance Check

We first run a covariate balance check. Meta Ads allows us to extract our data on impressions and clicks with either 1) gender and age or 2) device type and placement. We first look at Tables 1 and 2 below to check covariate balance for gender and age respectively.

Table 1: Covariate Balance Check for Gender

Gender	Control_Prop	Treatment_Prop	p_value
female	0.5387	0.5359	0.5984
male	0.4487	0.4549	0.2353
unknown	0.0125	0.0091	0.0019

Table 2: Covariate Balance Check for Age Buckets

Age	Control_Prop	Treatment_Prop	p_value
18-24	0.1799	0.2794	0
25-34	0.1791	0.2313	0
35-44	0.1655	0.1355	0
45-54	0.1320	0.1111	0
55-64	0.1507	0.1107	0
65+	0.1928	0.1320	0

Regarding gender, we note that we do have covariate balance for Male and Female. However, we do not have covariate balance for Unknown. Unknown could refer to the small percentage of users that were using company accounts, or did not input their gender as Male or Female in their Meta account. It is not dire, given that the proportion of users with unknown gender is extremely small, but we would want to control for gender.

We also note that we do not have covariate balance for age. This is likely due to the Meta Ads platform's targeting algorithm, which although we tried to mitigate by clustering on ZIP code which would technically allow the same algorithm to be applied to both treatment and control ZIP codes, the algorithm might have interpreted both ads differently and thus targeted different age groups with them. We see that age groups 18-24 and 25-34 were much more likely to be targeted with the treatment ad than the control ad, and age groups 55-64 and 65+ were much more likely to be targeted with the control ad than the treatment ad. This is unfortunate, and we thus will need to control for age as well.

We now look at Tables 3 and 4 to check covariate balance for device and placement respectively.

Table 3: Covariate Balance Check for Device Categories

Device	Control_Prop	Treatment_Prop	p_value
iphone	0.7131	0.7985	0.0000
android_smartphone	0.2525	0.1687	0.0000
ipad	0.0081	0.0085	0.7171
android_tablet	0.0018	0.0027	0.0754
desktop	0.0241	0.0212	0.0679
other	0.0004	0.0004	1.0000
ipod	0.0000	0.0001	0.4058

Table 4: Covariate Balance Check for Placement Categories

Placement	Control_Prop	Treatment_Prop	p_value
Instagram Stories	0.3060	0.5376	0.0000
Feed	0.1920	0.1501	0.0000
Facebook Stories	0.0506	0.0305	0.0000
In-stream video	0.3626	0.2156	0.0000
Instagram Reels	0.0441	0.0287	0.0000
Marketplace	0.0164	0.0133	0.0181
Right column	0.0226	0.0200	0.0939
Explore	0.0021	0.0014	0.1214
Facebook profile feed	0.0013	0.0014	0.9774
Search results	0.0024	0.0014	0.0258

Critically, we observe that we do not have covariate balance for certain device categories such as **iphone** and **android_smartphone**. We also observe that we do not have covariate balance for certain placement categories like Instagram Stories, Feed, among others. Once again, this is likely because the Meta Ads targeting algorithm treated both ads differently. To correct for this, we should thus control for both device and placement categories.

3.2 Model Building and Discussion

We now build three models to observe if we get any average treatment effect.

1. Model 1 Baseline: We regress the click through rate **CTR (all)** on the treatment assignment. As we have ZIP code level data, we use robust standard errors.
2. Model 2: Similar to Model 1, but we add in controls for age and gender. As we have multiple observations per ZIP code (with hierarchical data with genders and ages for each ZIP code), we use robust standard errors clustered on ZIP code.
3. Model 3: Similar to Model 1, but we add in controls for device and placement. As we have multiple observations per ZIP code (with hierarchical data with device and placement for each ZIP code), we use robust standard errors clustered on ZIP code.

The three models are displayed in Table 5. We first note that we do not have a statistically significant coefficient on the treatment assignment for Model 1. Interestingly, models 2 and 3 have the same coefficient for the treatment assignment. This is unexpected, but also reassuring that none of the models is uniquely better or different than the other. However, due to covariate imbalance, we are still unable to claim that the coefficient is unbiased, as we might still have unobserved heterogeneity.

Generally, regardless of control or treatment, we see that users of the age groups **Age55-64** and **Age65+** are more likely to click, when compared to the baseline of **Age18-24**. We also note that users of the Gender **male** are less likely to click when compared to the baseline of Gender **female**.

We also note that all Placements obtained a statistically significantly lower click through rate than the reference level of **Facebook Stories**.

We also ran a model that included interaction terms of treatment with age and gender; there was nothing statistically significant found.

Finally, we ran a model, Model 4, that included interaction terms of treatment with device and placement.

4. Model 4: Similar to Model 3, but with interaction terms between the treatment assignment variable and device and placement. The results are in Table 6.

Interestingly, from Table 6, we note that users using an iPad were less likely to click on the treatment ad. However, when we observe the covariate balance for iPad in Table 3, we do see covariate balance between

control and treatment groups. This is unexpected, since we would have expected the Meta algorithm to perhaps favor users with the iPad in the control group, but that is not what we see, suggesting that perhaps Meta's algorithm did not account for this discrepancy.

Table 5: Regression Results For Models 1-3

	<i>Dependent variable:</i>		
	(1)	(2)	(3)
AssignmentTreatment	-0.124 (0.108)	-0.063 (0.105)	-0.063 (0.102)
Age25-34		0.211* (0.123)	
Age35-44		0.170 (0.122)	
Age45-54		0.200 (0.141)	
Age55-64		0.392*** (0.141)	
Age65+		0.742*** (0.176)	
Gendermale		-0.273*** (0.086)	
Genderunknown		0.073 (0.478)	
'Impression device'android_tablet			2.460 (2.674)
'Impression device'desktop			2.582 (2.149)
'Impression device'ipad			0.304 (0.598)
'Impression device'iphone			-0.059 (0.114)
'Impression device'ipod			-1.005*** (0.315)
'Impression device'other			-1.359*** (0.268)
PlacementExplore			-2.479*** (0.405)
PlacementFacebook profile feed			-2.452*** (0.397)
PlacementFeed			-1.011*** (0.383)
PlacementIn-stream video			-1.942*** (0.405)
PlacementInstagram Reels			-1.943*** (0.465)
PlacementInstagram Stories			-1.871*** (0.403)
PlacementMarketplace			-2.381*** (0.491)
PlacementRight column			-5.064** (2.166)
PlacementSearch results			-2.484*** (0.406)
Constant	0.811*** (0.082)	0.638*** (0.102)	2.509*** (0.418)
Observations	60	849	1,007
R ²	0.020	0.029	0.078
Adjusted R ²	0.003	0.020	0.063
Residual Std. Error	10.950 (df = 58)	10.726 (df = 840)	11.456 (df = 990)
F Statistic	1.186 (df = 1; 58)	3.138*** (df = 8; 840)	5.217*** (df = 16; 990)

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 6: Regression Results For Model 4

	<i>Dependent variable:</i>
	‘CTR (all)’
AssignmentTreatment	-0.244 (0.809)
‘Impression device’android_tablet	4.340 (5.299)
‘Impression device’desktop	1.922 (1.898)
‘Impression device’ipad	1.299 (1.088)
‘Impression device’iphone	0.082 (0.142)
‘Impression device’ipod	-1.226*** (0.320)
‘Impression device’other	-1.846*** (0.357)
PlacementExplore	-2.587*** (0.532)
PlacementFacebook profile feed	-2.597*** (0.532)
PlacementFeed	-1.157** (0.515)
PlacementIn-stream video	-2.102*** (0.542)
PlacementInstagram Reels	-2.122*** (0.625)
PlacementInstagram Stories	-2.097*** (0.545)
PlacementMarketplace	-2.666*** (0.585)
PlacementRight column	-4.491** (2.054)
PlacementSearch results	-2.621*** (0.523)
AssignmentTreatment:‘Impression device’android_tablet	-3.463 (5.738)
AssignmentTreatment:‘Impression device’desktop	1.586 (5.150)
AssignmentTreatment:‘Impression device’ipad	-2.207** (1.104)
AssignmentTreatment:‘Impression device’iphone	-0.388 (0.243)
AssignmentTreatment:‘Impression device’ipod	
AssignmentTreatment:‘Impression device’other	0.942 (0.441)
AssignmentTreatment:PlacementExplore	0.367 (0.790)
AssignmentTreatment:PlacementFacebook profile feed	0.458 (0.763)
AssignmentTreatment:PlacementFeed	0.442 (0.738)
AssignmentTreatment:PlacementIn-stream video	0.436 (0.780)
AssignmentTreatment:PlacementInstagram Reels	0.484 (0.889)
AssignmentTreatment:PlacementInstagram Stories	0.613 (0.784)
AssignmentTreatment:PlacementMarketplace	0.768 (1.011)
AssignmentTreatment:PlacementRight column	-1.342* (5.110)
AssignmentTreatment:PlacementSearch results	0.484 (0.774)
Constant	2.569*** (0.542)
Observations	1,007
R ²	0.083
Adjusted R ²	0.055
Residual Std. Error	11.502 (df = 976)
F Statistic	2.963*** (df = 30; 976)
Note:	*p<0.1; **p<0.05; ***p<0.01

4 Conclusion and Next Steps

The randomized experiment aimed to identify the causal effects of using specific, quantifiable messaging to help organizations better allocate their fundraising budgets. While prior research has suggested that messaging strategies can significantly influence donation rates and amounts, our study found no statistically significant differences based on the type of advertisement users were exposed to.

To strengthen future iterations of this experiment, we could move beyond Meta’s platform for user covariate data by finding another platform to collect this information uniformly and directly, to better ensure covariate balance between treatment and control groups.

Additionally, we could also consider building the technical infrastructure in future iterations into SYRCL’s donation platform to measure real donation counts and amounts. This will allow us to capture conversion rates, average gift size, and total revenue—outcomes we were unable to assess in this pilot. In this particular experiment, however, this was irrelevant as there were no donations from either treatment or control ads.

Finally, we could expand the variety and quality of advertisements tested in a multi-factorial design. Currently, each user is exposed to only one non-randomized ad, and a single outcome is measured. In future studies, we could introduce multiple versions of both quantifiable and general messages and expose the same group of users to each. This approach would help control for variability in ad design quality and provide more robust insights into the effectiveness of different messaging strategies.

5 Reference

Hattaway Communications. (2023, May). How and why we give: Research insights on the aspirations and motivations that inspire people to give and volunteer. The Generosity Commission. <https://www.thegenerositycommission.org/wp-content/uploads/2023/05/Hattaway-Generosity-Insights-Report-5-10-23-FINAL.pdf>